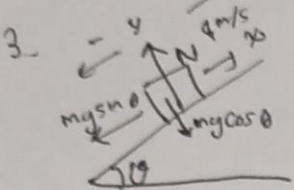


TUGAS 3 FISIKA 1 Modul 03

Nomor 3, 11, 13, 14 dan 15

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 Tanggal : 14 Oktober 2025

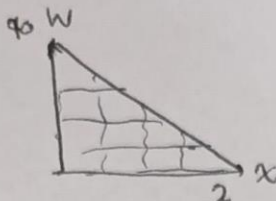
Jawaban :



$$K_s = \frac{1}{2} m v_0^2$$

$$\Rightarrow 40 = \frac{1}{2} m \cdot 4^2$$

$$\Rightarrow m = 5 \text{ kg}$$



$$dW = F \cdot dx$$

$$\Rightarrow \frac{dW}{dx} = F$$

$$\Rightarrow F = \text{gradien grafik}$$

$F = \frac{0 - 40}{2 - 0} = -20 \text{ N}$. Gaya yang bekerja pada sumbu-x hanya $-mg \sin \theta$ sehingga (negatif karena berlawanan arah)

$$\Rightarrow F = -mg \sin \theta$$

$$\Rightarrow -20 = -5 \cdot 9,8 \cdot \sin \theta$$

$$\Rightarrow \sin \theta = \frac{20}{49}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$= \sqrt{1 - \left(\frac{20}{49}\right)^2}$$

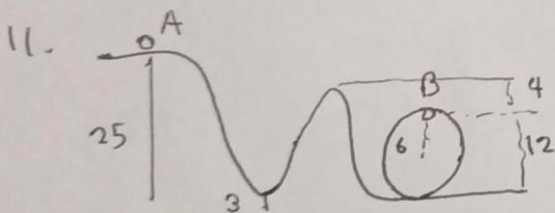
$$= \sqrt{\frac{2001}{2401}} = \frac{1}{49} \sqrt{2001}$$

$N - mg \cos \theta = 0$ (Hukum I Newton)

$$\Rightarrow N = mg \cos \theta$$

$$= 5 \cdot 9,8 \cdot \frac{1}{49} \sqrt{2001}$$

$$= \sqrt{2001} \text{ N} \approx 44,7325 \text{ N}$$



$m = 350 \text{ kg}$, licin $\Rightarrow E_{mA} = E_{mB} = E_{\text{mekanik}}$, $v_A = 0 \text{ m/s}$

Menggunakan Hukum Kekekalan Mekanik

$$E_{m1} = E_{m2} \Rightarrow E_{k1} + E_{p1} = E_{k2} + E_{p2}$$

$$E_{mA} = mgh_A + \frac{1}{2} m v_A^2$$

$$= 350 \cdot 9,8 \cdot 25$$

$$= 85750 \text{ J}$$

(a) v_B ? $E_{mA} = E_{mB}$

$$\Rightarrow 85750 = mgh_B + \frac{1}{2} m v_B^2$$

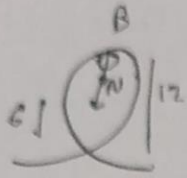
$$= 350 \cdot 9,8 \cdot 12 + \frac{1}{2} \cdot 350 \cdot v_B^2$$

$$= 41160 + 175 v_B^2$$

$$\Rightarrow v_B^2 \cdot 175 = 44590$$

$$\Rightarrow v_B^2 = \frac{1274}{5} \Rightarrow v_B = \frac{7}{5} \sqrt{130} \text{ m/s} \approx 15,9624 \text{ m/s}$$

(b)



Setiap gerak melingkar pasti memiliki gaya sentripetal dan sesuai dengan gambar, gaya kontak ini sama persis dengan gaya sentripetal sehingga

$$N = \frac{mv^2}{r} \Rightarrow N_0 = \frac{mv_0^2}{r} \text{ dengan } r = 6 \text{ m}$$

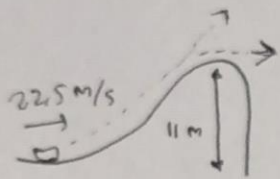
$$v_0 = \frac{7}{5} \sqrt{130} \text{ m/s}$$

$$\text{Maka, } N = \frac{350 \cdot \left(\frac{7}{5} \sqrt{130}\right)^2}{6}$$

$$= \frac{350 \cdot \frac{49}{25} \cdot 130}{6}$$

$$= \boxed{22295 \text{ N}}$$

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$m = 125 \text{ kg}$ Pakai Hukum Kekekalan energi
licin

$$y = v_y t + \frac{1}{2} g t^2$$

$$11 = 4.9 t^2$$

$$\Rightarrow t = \sqrt{\frac{11}{4.9}} = \frac{1}{7} \sqrt{110}$$

$$E_{\text{meo}} = E_{\text{mei}}$$

$$mgh_0 + \frac{1}{2} m v_0^2 = mgh_1 + \frac{1}{2} m v_1^2$$

$$\Rightarrow gh_0 + \frac{1}{2} v_0^2 = gh_1 + \frac{1}{2} v_1^2$$

$$\Rightarrow \frac{1}{2} (22.5)^2 = 9.8 \cdot 11 + \frac{1}{2} v_1^2$$

$$\Rightarrow \frac{1}{2} v_1^2 = 145.325$$

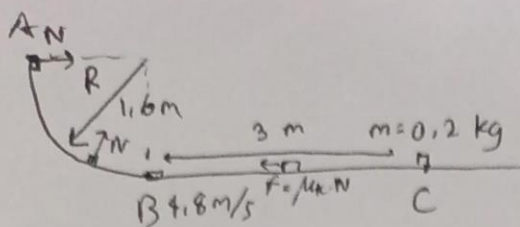
$$\Rightarrow v_1 = \sqrt{290.65} \text{ [ini } v_x \text{ karena bentuknya } \sqrt{\quad}]$$

$$\Delta x = v \cdot \Delta t$$

$$= \sqrt{290.65} \cdot \frac{1}{7} \sqrt{110}$$

$$= \boxed{\frac{1}{7} \sqrt{31971.5} \text{ m} \approx 25.59368 \text{ m}}$$

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$$m = 0,2 \text{ kg}$$

$$G_{PB} = E_{PC} = 0 \text{ J}$$

$$(a) v_B = 4,8 \text{ m/s}, v_C = 0$$

$$N = mg = 0,2 \cdot 9,8 = 1,96 \text{ N}$$

$$E_{kb} - E_{kf} = E_{kc}$$

$$\Rightarrow \frac{1}{2}mv_b^2 - F \cdot \Delta x = 0$$

$$f = \mu_k \cdot N$$

$$\Rightarrow F \cdot 3 = \frac{1}{2} \cdot 0,2 \cdot 4,8^2$$

$$\Rightarrow \mu_k = \frac{f}{N}$$

$$\Rightarrow F = \frac{2,304}{3} \text{ N}$$

$$= \frac{2,304}{1,96}$$

$$\Rightarrow f = \frac{2,304}{3} \text{ N}$$

$$= \frac{2,304}{1,96} \approx 1,1755 \text{ N}$$

$$E_{mb} - E_{ma} = \text{akhir-awal} \quad 2$$

$$(b) W_f = -(E_{ma} - E_{mb})$$

$$= -\left(\left(\frac{1}{2}mv_a^2 + mgh_a\right) - \left(\frac{1}{2}mv_b^2 + mgh_b\right)\right)$$

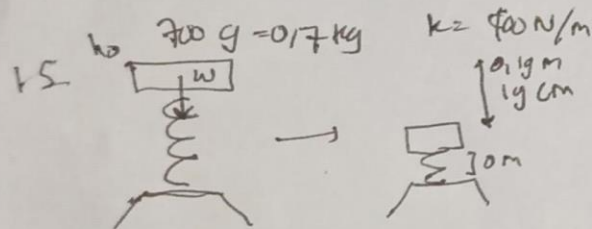
$$= -(mgh_a - \frac{1}{2}mv_b^2)$$

$$\Rightarrow 0,2(9,8 \cdot 1,6 - \frac{1}{2} \cdot (4,8)^2)$$

$$\Rightarrow 0,2(4,16)$$

$$\Rightarrow E_{ma} - E_{mb}$$

$$= \boxed{-0,832 \text{ J}} \quad (\text{atau } +0,832 \text{ J karena mengurangi energi})$$



$$W = mg = 0,7 \cdot 9,8 = 6,86 \text{ N}$$

$$(a) W_{kp} = \frac{1}{2}k\Delta x^2 = \frac{1}{2} \cdot 400 \cdot (0,19)^2 = 200 \cdot (0,0361) = \boxed{7,22 \text{ J}}$$

(b) Menggunakan Hukum III Newton,

didapat $W_{pt} = -W_{kp}$

$$= \boxed{-7,22 \text{ J}}$$

$$(c) E_{m0} = E_{m1}$$

$$\Rightarrow mg(h_0 + \Delta x) = \frac{1}{2}k\Delta x^2$$

$$\Rightarrow 0,7 \cdot 9,8(h_0 + 0,19) = 7,22$$

$$\Rightarrow h_0 + 0,19 = \frac{7,22}{6,86}$$

$$\Rightarrow h_0 = \frac{7,22}{6,86} - 0,19 \text{ m} \approx 0,8624 \text{ m} \text{ atau } 86,24 \text{ cm}$$

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(d) Menggunakan Hukum Kekekalan Energi,

$$E_{m_0} = E_{m_1}$$

$$\Rightarrow mg(2h_0 + \Delta x') = \frac{1}{2} k (\Delta x')^2$$

$$\Rightarrow 2h_0 + \Delta x' = \frac{200 \Delta x'^2}{6,86}$$

$$\Rightarrow \frac{200}{6,86} \Delta x'^2 - \Delta x' - 2h_0 = 0$$

$$\Rightarrow \frac{200}{6,86} \Delta x'^2 - \Delta x' - 2\left(\frac{7,22}{6,86} - 0,19\right) = 0$$

Gunakan rumus ABC dengan $a = \frac{200}{6,86}$, $b = -1$, dan $c = \left(0,19 - \frac{7,22}{6,86}\right) \cdot 2$

$$\Delta x' = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{200}{6,86} \left(0,19 - \frac{7,22}{6,86}\right) \cdot 2}}{2 \cdot \frac{200}{6,86}}$$

= (Ambil akar positif $\Rightarrow \pm \rightarrow +$)

$$= \frac{1 + \sqrt{1 + \frac{800}{6,86} \left(\frac{7,22}{6,86} - 0,19\right) \cdot 2}}{\frac{200}{6,86}}$$

$$\approx \boxed{0,260994 \text{ m} = 26,0994 \text{ cm}}$$